

TransMLA: Multi-head Latent Attention

Abstract: TransMLA addresses KV-cache compression via multi-head latent attention.

Section 3.2 RoPE Compatibility:

A RoPE format mismatch causes incorrect positional encoding when converting from MHA to MLA. The mismatch degrades model quality and must be fixed before deployment.

Results: 16x KV-cache reduction on SmolLM and Qwen3-4B benchmarks with less than 0.5 percent degrad